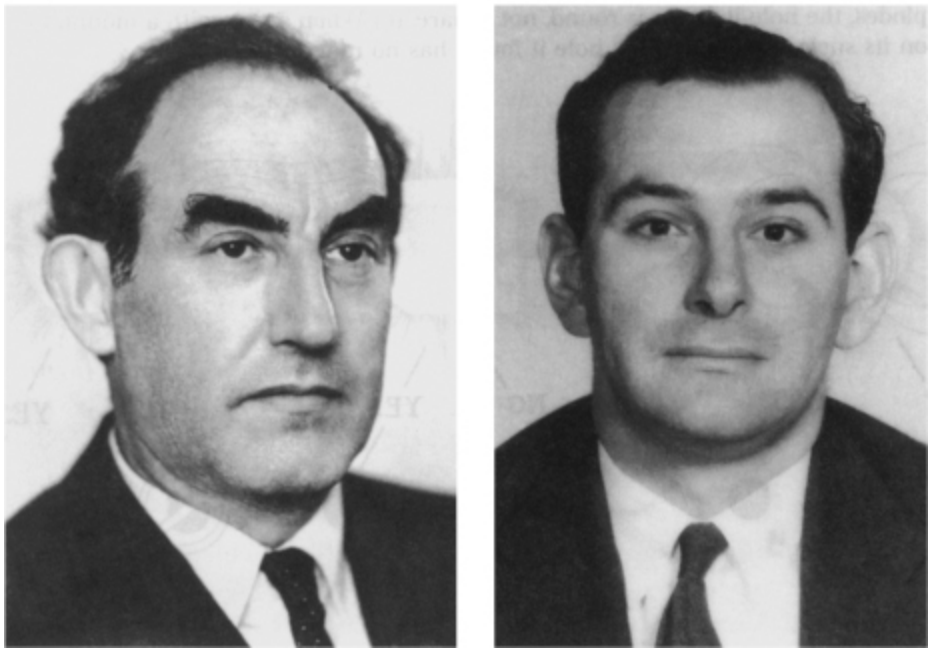
the second sort. It grew out of research by the intellectual progeny of all three great mentors, Zel'dovich, Wheeler, and Sciama, and out of research by many others. In the following pages, we shall watch as this myriad of researchers struggles step by step, bit by bit, to formulate the concept of a black hole's hairlessness, prove it, and grasp its implications.

The first hints that "a black hole has no hair" came in 1964, from Vitaly Lazarevich Ginzburg, the man who had invented the LiD fuel for the Soviet hydrogen bomb, and whose wife's alleged complicity in a plot to kill Stalin had freed him from further bomb design work (Chapter 6). Astronomers at Caltech had just discovered *quasars,* enigmatic, explosive objects in the most distant reaches of the Universe, and Ginzburg was trying to understand how quasars might be powered (Chapter 9). One possibility, Ginzburg thought, might be the implosion of a magnetized, supermassive star to form a black hole. The magnetic field lines of such a star would have the shape shown in the upper part of Figure 7.3a—the same shape as the Earth's magnetic field lines. As the star implodes, its field lines might become strongly compressed and then explode violently, releasing huge energy, Ginzburg speculated; and this might help to explain quasars.



Left: **Vitaly Lazarevich Ginzburg (ca. 1962), the person who produced the first evidence for the "no-hair conjecture."** ***Right:*** **Werner Israel (in 1964), the person who devised the first rigorous proof that the "no-hair**